

thisdogdoesnotexist.com

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Abstract—This project includes the development of a Generative Adversarial Network to create realistic, artificial images of dogs. Inference will be served and accessible through a simple web UI.

I. PROBLEM STATEMENT

Generative AI for images is a popular area in the current age of AI. Through this project, we will study and explore how to create a realistic 64x64 RGB image of a dog using a Generative Adversarial Network. Dogs have a much less uniform alignment of face and bodies than human faces do, making this project more technically challenging than a similar project in the realm of human face generation. Similarly to the project inspiration, <https://thispersondoesnotexist.com>, an image of a dog, generated from an instance of inference from our model, will be displayed when visiting the web UI.

II. DATASET

To reach a sufficient amount of data, we will collate three unique data sets into one. All of the three data sets are free and available right now.

Dataset	Images
https://cg.cs.tsinghua.edu.cn/ThuDogs/	70,428
https://www.robots.ox.ac.uk/~vgg/data/pets/	4,978
https://data.mendeley.com/datasets/v5j6m8dzhv/1	1,657
Total	77,063

III. NEURAL APPROACH

We will train two different deep learning models: a Variational Autoencoder [1] will serve as the baseline, and a Generative Adversarial Network [2] will be the frontier. This combination works well since Variational Autoencoders, intrinsically, produce more stable but blurrier images at inference when compared to a Generative Adversarial Network trained on the same image set.

IV. SUCCESS EVALUATION

We will evaluate success of the models by using The Fréchet Inception Distance. More robust than subjective human inspection, The Fréchet Inception Distance numerically measures how close an artificially generated set of images is to the images in the training set [3]. We will take a large sample of n generated images from each model, and calculate the Fréchet Inception Distance from each model's generated set to the training set.

V. FEASIBILITY

By pruning the image quality of our data set to 64x64, this project is feasible within the time constraint of one semester and the compute constraint of Google Colab. Other experiments training a Generative Adversarial Network on Google Colab indicate that 139 epochs of a 28x28 grayscale data set of 70,000 images took two hours [4]. Extrapolating this to our model, the estimated lower bound training time for our 77,063 image data set is 15 minutes per epoch. A full run of 100 to 200 epochs is therefore in the estimated range of 25 to 50 hours per model, which exceeds a single Colab session of 12 hours, so we will checkpoint model weights to Google Drive and resume across sessions. If further pruning is needed, halving the image quality to 32x32 would reduce the training time by 75%, or halving the image count of the training set would halve the training time as well. Taking both pruning measures would reduce the training time by a total of 87%.

REFERENCES

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